RAMRUP SATPATI

DATA SCIENCE / MACHINE LEARNING / SOFTWARE ENGINEERING



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IIT Madras • Chennai, India

IIT Madras BS student in Data Science and Applications with project experience spanning machine learning, deep learning, generative AI, mathematical optimization, and full-stack applications. Builds reproducible analytical workflows and practical software systems, combining model development, data analysis, and structured project execution.

EDUCATION

Indian Institute of Technology Madras (IIT Madras) — **2024 – Present**
BS in Data Science and Applications · Degree Level — Chennai, India

CGPA: 7.57 / 10.0 • Top Course Grade – Diploma Honor Badge

Completed theoretical components for dual Diplomas in Programming and Data Science. Coursework includes Machine Learning Foundations & Practice, Deep Learning & GenAI (Viva: 45/50), Business Data Management, Modern App Development I & II, Tools in Data Science, and Systems & Usable Security (81%, Elite Distinction).

TECHNICAL SKILLS

Languages & Data	Python, SQL (PostgreSQL, SQLite), JavaScript, Bash/Shell, HTML5/CSS3
ML / AI	PyTorch, TensorFlow, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, transformer models
Analytics & Optimization	EDA, feature engineering, model evaluation, constrained linear programming (Simplex / SciPy), CLV, econometrics
Development & Tools	Flask, Vue.js, Streamlit, REST APIs, Git/GitHub, Docker, Linux/macOS, Jupyter Notebooks

SELECTED PROJECTS

Sales & Service Mix Optimization — BDM Capstone — Oct 2026
github.com/24f3004027/Business_Data_Management
- Analyzed a primary field survey of 220 retail/commercial clients to evaluate after-sales service (AMC) attachment economics.
- Built a constrained Simplex linear-programming model under technician-labor and working-capital limits; identified an optimal mix of 74 sales units and 132 AMC contracts/month, with a reported 28.4% profit increase (■1.226M/month) and 3.0× return on technician payroll.

Deep Learning & Generative AI Systems — 2026
github.com/24f3004027/Deep_Learning_Project
- Implemented deep-learning pipelines using CNNs, recurrent sequence models, and transformer representations in PyTorch/TensorFlow.
- Applied transfer learning and latent-space representations; scored 45/50 (90%) in viva and 10/10 on the technical report.

Placement Portal Application V2 — MAD 2 Capstone — 2025–2026
github.com/24f3004027/Placement_Portal_Application_V2
- Developed a placement-management platform with role-based access control for administrators, corporate recruiters, and candidates.
- Built Flask REST APIs and a Vue.js interface; project description also lists Celery/Redis background-job queues.

Vehicle Parking Management System — MAD 1 2025

github.com/24f3004027/vehicle-parking-appv1

- Built a multi-slot parking application with ticket timestamps, dynamic billing, SQLite storage, server-side validation, and session security using Python/Flask.

Hospital Management System — HMS 2025

github.com/24f3004027/HMS-MAD-Projectv1

- Designed a clinical records and appointment system for doctors, patients, and administrators, including relational schemas, appointment-slot booking, and diagnostic summaries.

Predictive ML Pipeline & Python Projects Monorepo 2024–2026

github.com/24f3004027/Machine_Learning_Practice_Project

- Built supervised classification/regression workflows with Scikit-learn and XGBoost, feature transformations, cross-validated grid search, and model-explainability metrics.
- Maintains a Python project repository of algorithmic tools, data structures, automation utilities, and numerical scripts: github.com/RSNPIIT/Python-Projects.

EXPERIENCE & LEADERSHIP

Open Source Contributor & Technical Project Lead · 2024–Present · Remote

Independent Data Science & Software Developer · GitHub: RSNPIIT

- Creates reproducible data-science and optimization repositories, documentation, and analytical codebooks; uses structured milestone tracking and collaborative problem-solving.

CERTIFICATIONS & LANGUAGES

Certifications: Certified in multiple courses through NPTEL (National Programme on Technology Enhanced Learning). Additional recognition: IIT Madras Top Course Grade ? Diploma Honor; Systems & Usable Security, Elite Distinction (81%).